

High Altitude Tiny Object Detection

VISUALIZATION OF THE RESULTS

Done by

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Filtered Tiny Image mAP

```
except:
    print(" (per-class breakdown not available)")

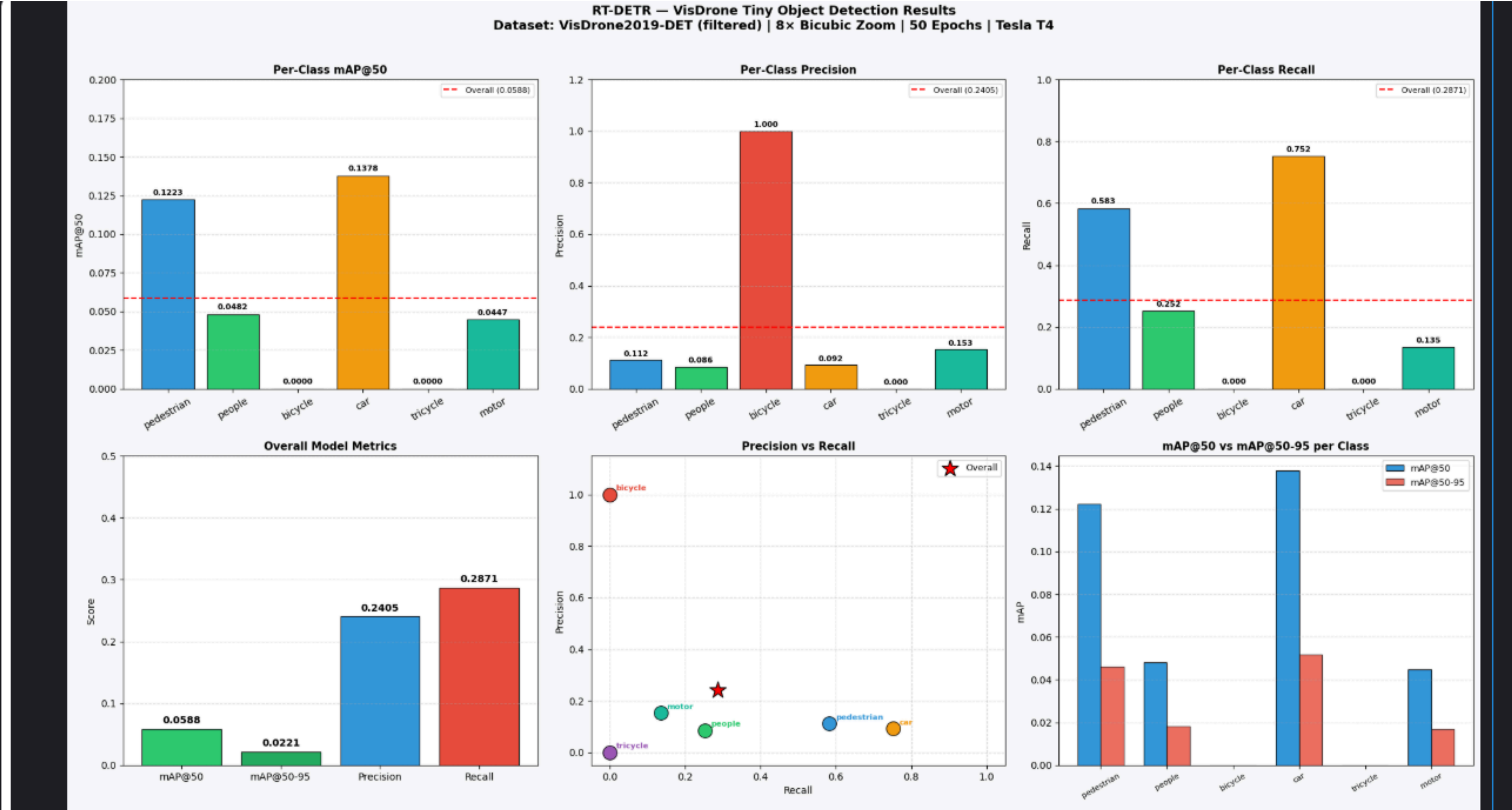
Ultralytics 8.4.53 🚀 Python-3.12.12 torch-2.10.0+cu128 CUDA:0 (Tesla T4, 14913MiB)
rt-detr-l summary: 310 layers, 31,996,070 parameters, 0 gradients, 103.5 GFLOPs
val: Fast image access ✅ (ping: 0.0±0.0 ms, read: 4117.4±141.3 MB/s, size: 5860.8 KB)
val: Scanning /kaggle/working/dataset_zoomed/val/labels.cache... 60 images, 0 backgrounds, 0 corrupt: 100% _____ 60/60 14.8Mit/s 0.0s
      Class      Images  Instances    Box(P   R    mAP50  mAP50-95): 100% _____ 15/15 1.1it/s 13.1s0.6s
      all         60       2500      0.24    0.287  0.0588   0.0221
    pedestrian    54       1152      0.112    0.583  0.122   0.0402
      people     50        468     0.0858   0.252  0.0482   0.0171
      bicycle    34         74         1         0         0         0
        car      37        492      0.092    0.752  0.138   0.0619
      tricycle   22         48         0         0         0         0
        motor    48        266      0.153    0.135  0.0447   0.0132

Speed: 2.7ms preprocess, 84.7ms inference, 0.0ms loss, 1.2ms postprocess per image
Results saved to /kaggle/working/runs/detect/val

=====
📊 RT-DETR mAP – VisDrone Tiny Objects 8× Zoom
=====
mAP@50      : 0.0588
mAP@50-95   : 0.0221
Precision   : 0.2405
Recall      : 0.2871
=====

Per-class mAP@50:
Class      mAP@50
-----
pedestrian 0.1223
people     0.0482
bicycle     0.0000
car         0.1378
tricycle    0.0000
motor       0.0447
```

Graphic Representation - Filtered Tiny Image mAP



RT DETR 8x Zoom Results

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RT-DETR RESULTS – VisDrone Tiny Object Detection						
8× Bicubic Zoom 50 Epochs Tesla T4 imgsz=640						
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Class	Images	Instances	P	R	mAP@50	mAP50-95

pedestrian	54	1152	0.1120	0.5830	0.1223	0.0402
people	50	468	0.0858	0.2520	0.0482	0.0171
bicycle	34	74	1.0000	0.0000	0.0000	0.0000
car	37	492	0.0920	0.7520	0.1378	0.0619
tricycle	22	48	0.0000	0.0000	0.0000	0.0000
motor	48	266	0.1530	0.1350	0.0447	0.0132

ALL	60	2500	0.2405	0.2871	0.0588	0.0221
=====						

KEY OBSERVATIONS:

- ✓

Car detection is strongest : mAP@50 = 0.1378
Recall = 0.752 (finds 75% of cars)
- ✓

Pedestrian works reasonably : mAP@50 = 0.1223
Recall = 0.583
- ⚠

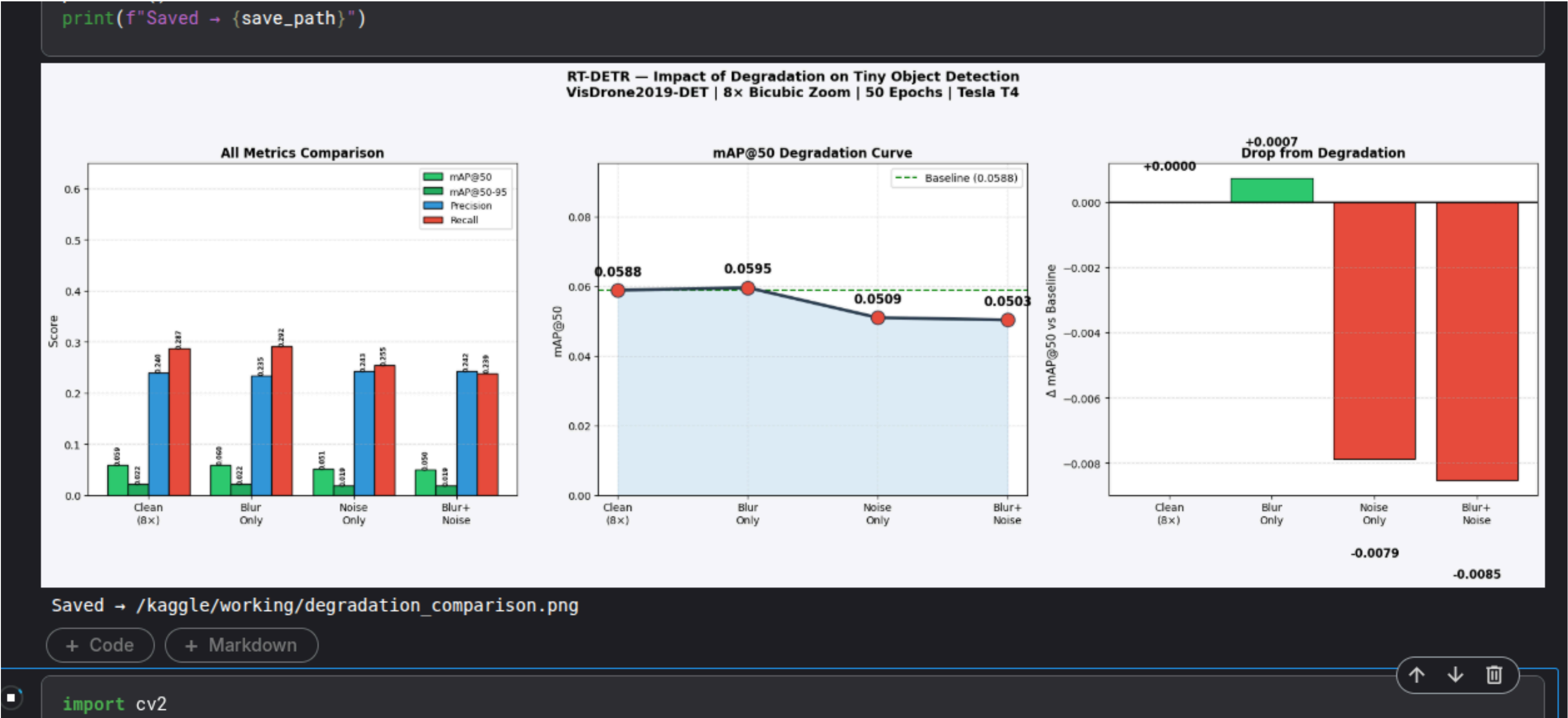
Bicycle mAP = 0 : Only 74 instances – too few to learn
- ⚠

Tricycle mAP = 0 : Only 48 instances – too few to learn
- ⚠

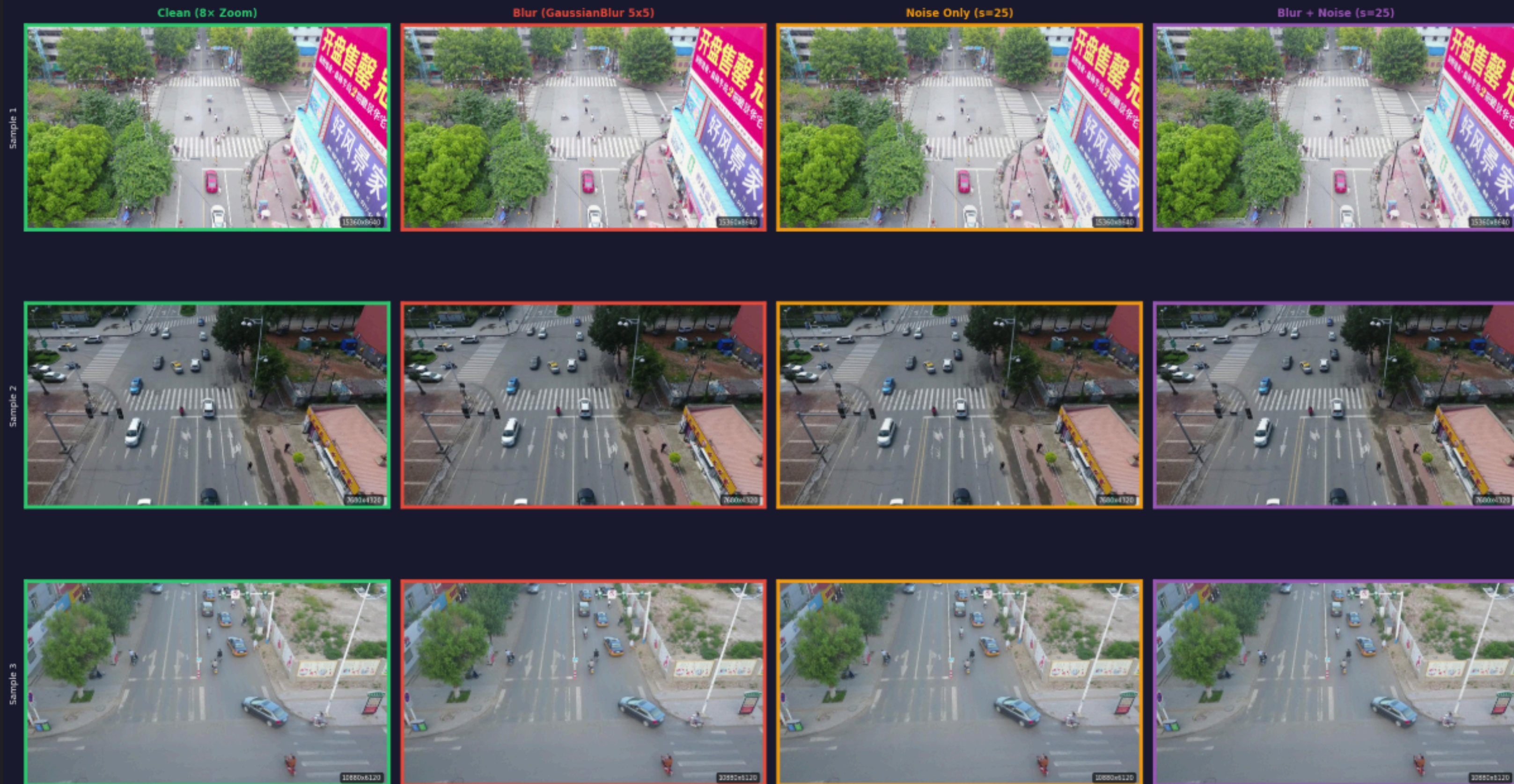
Low overall mAP (0.0588) : Expected at 6km altitude tiny objects
- 💡

Next step : Apply SAHI for +5-15 mAP improvement

Graphic Representation - RTDETR 8x Zoom, Blur Only, Noise Only, Blur + Noise



Degradation Conditions — 6km Altitude Simulation
VisDrone2019-DET | 8x Bicubic Zoom | RT-DETR Evaluation



Most object-dense image: 0000086_01954_d_0000005 (173 objects)

Image size: 7680 × 4320

Crop region: (1444,139) → (3364,1219)

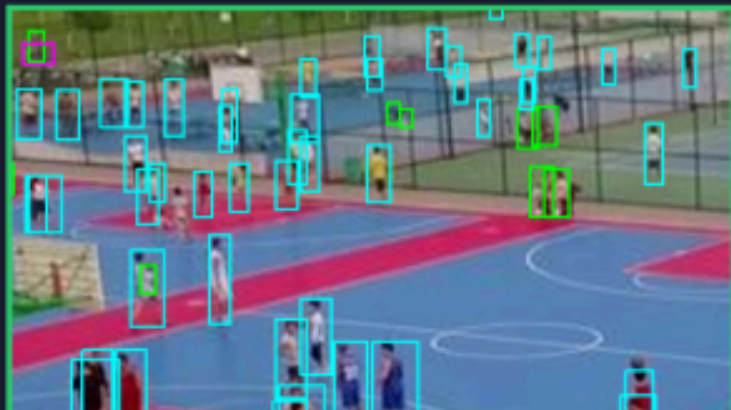
Crop size : 1920 × 1080

Tiny Object Degradation — Zoomed Crop View
Same Region | Same Model | 6km Altitude Conditions

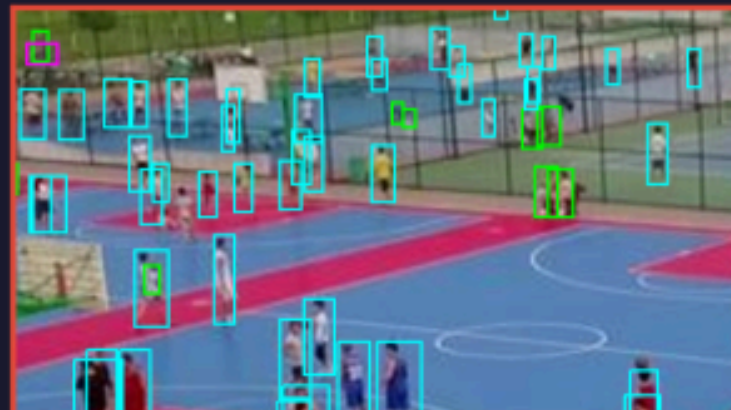
Full Image — 0000086_01954_d_0000005 (7680×4320px) | 173 objects | Yellow box = crop region



Clean (8× Zoom)



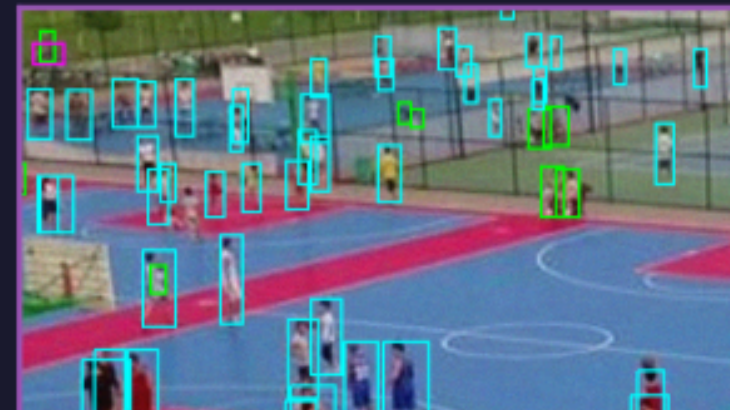
Blur (GaussianBlur 5×5)



Noise Only ($\sigma=25$)



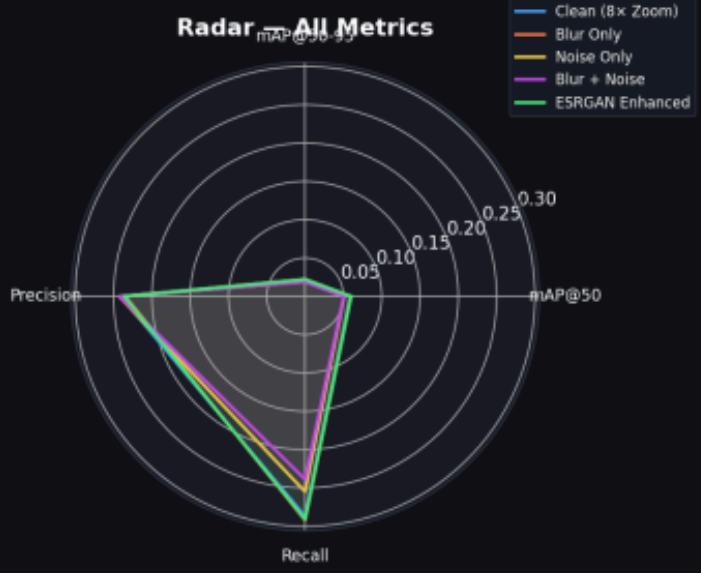
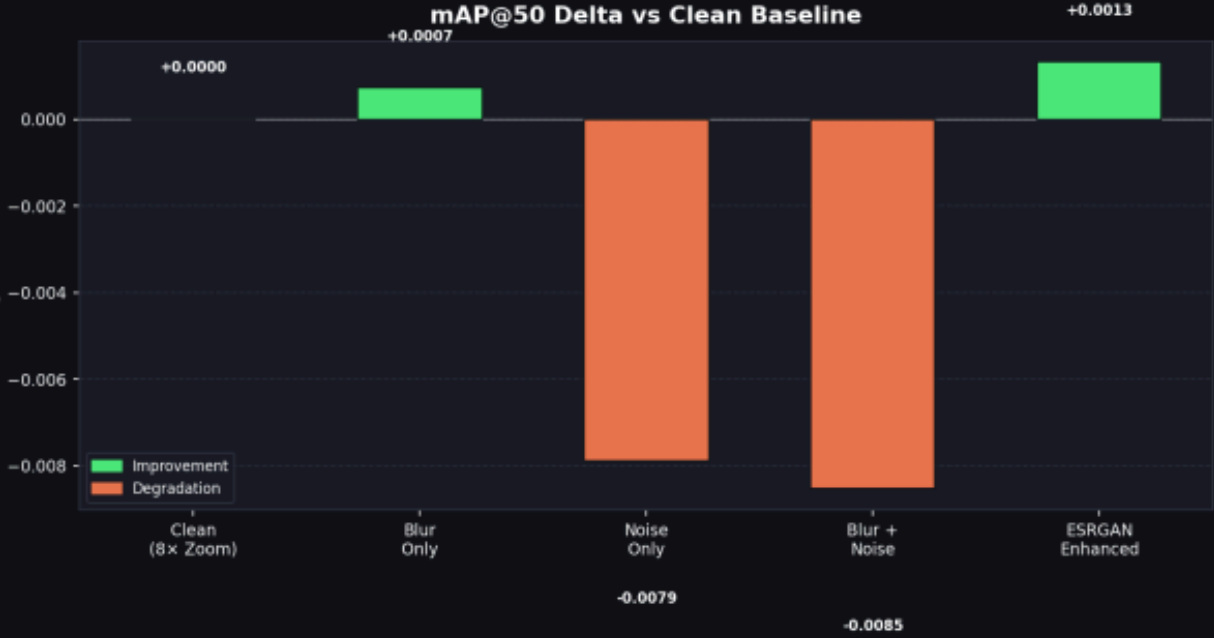
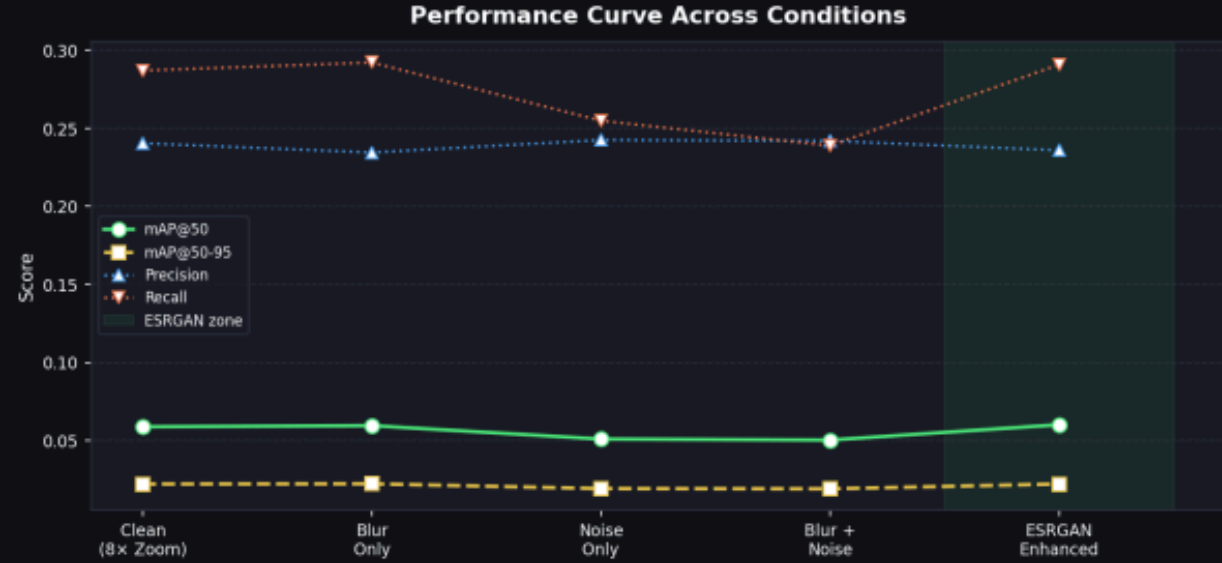
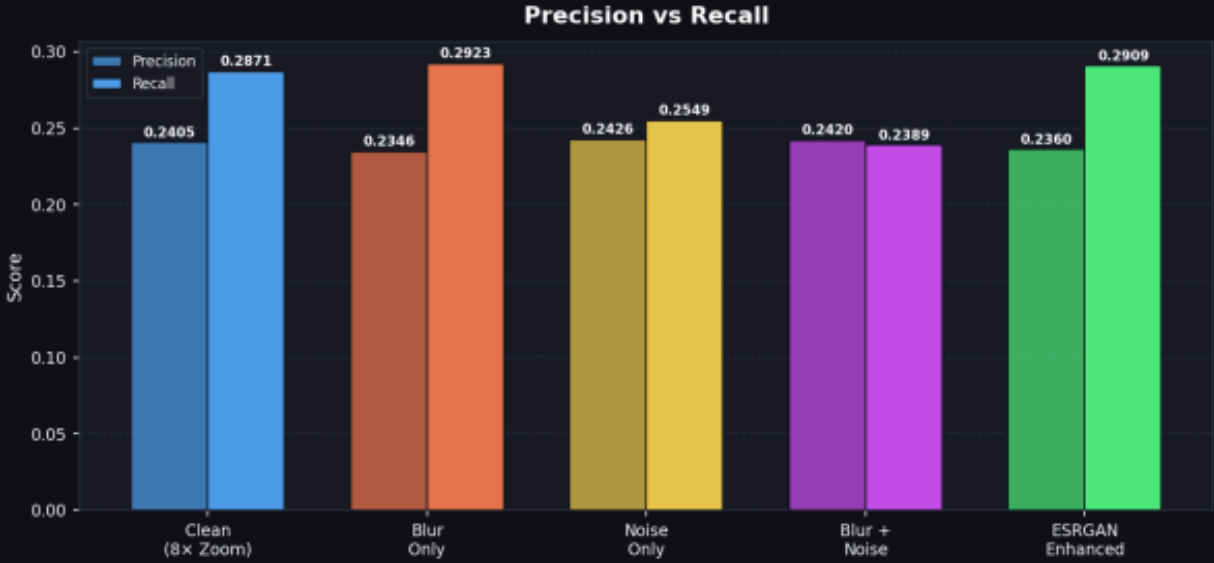
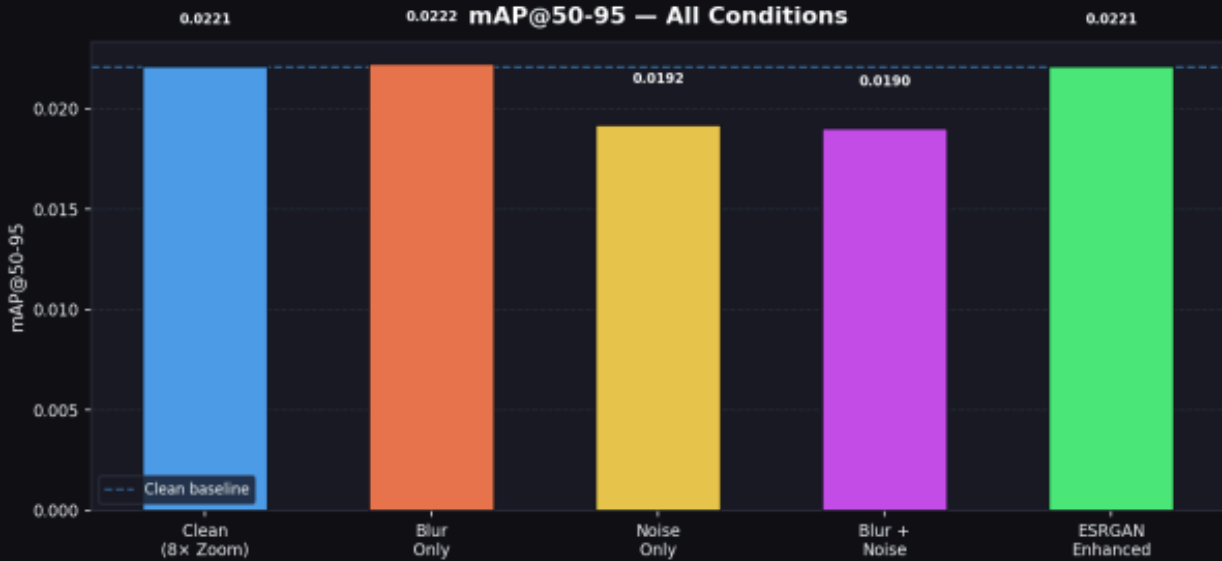
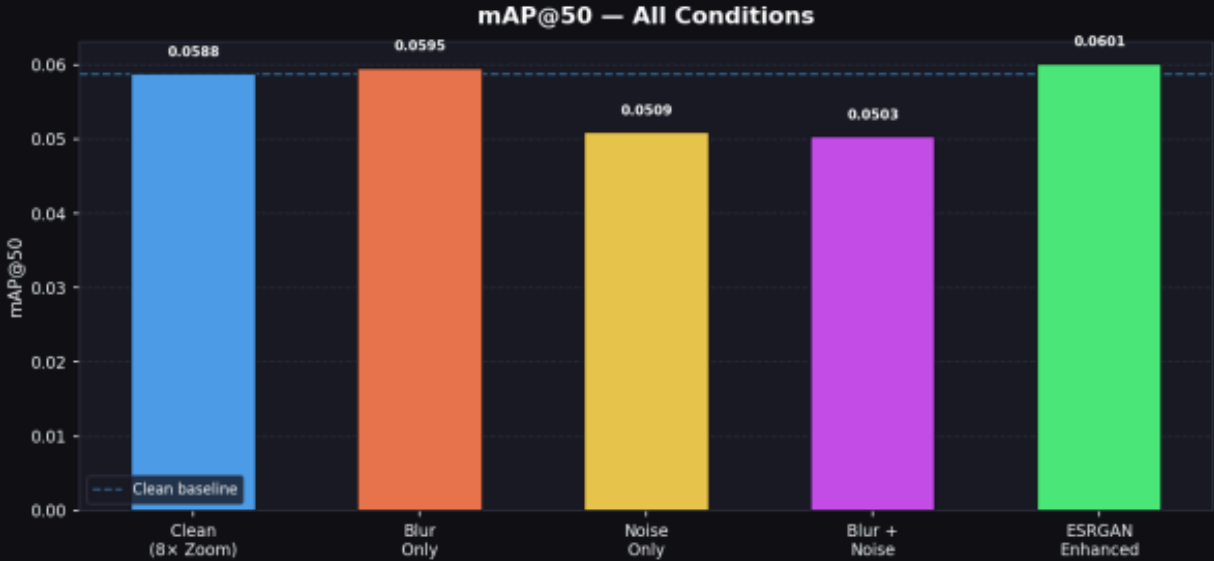
Blur + Noise ($\sigma=25$)



ESRGAN Enhanced Comparision

COMPLETE RESULTS – ALL 5 CONDITIONS					
Condition	mAP@50	mAP50-95	Precision	Recall	Δ mAP@50
Clean (8× Zoom)	0.0588	0.0221	0.2405	0.2871	baseline
Blur Only	0.0595	0.0222	0.2346	0.2923	+0.0007
Noise Only	0.0509	0.0192	0.2426	0.2549	-0.0079
Blur+Noise	0.0503	0.0190	0.2420	0.2389	-0.0085
ESRGAN Enhanced	0.0601	0.0221	0.2360	0.2909	+0.0013

RT-DETR · VisDrone Tiny Object Detection
Clean vs Blur vs Noise vs Blur+Noise vs ESRGAN Enhanced



ESRGAN + SAHI mAP

Computing mAP: 100%|██████████| 50/50 [00:00<00:00, 74.88it/s]

SAHI + ESRGAN metrics:

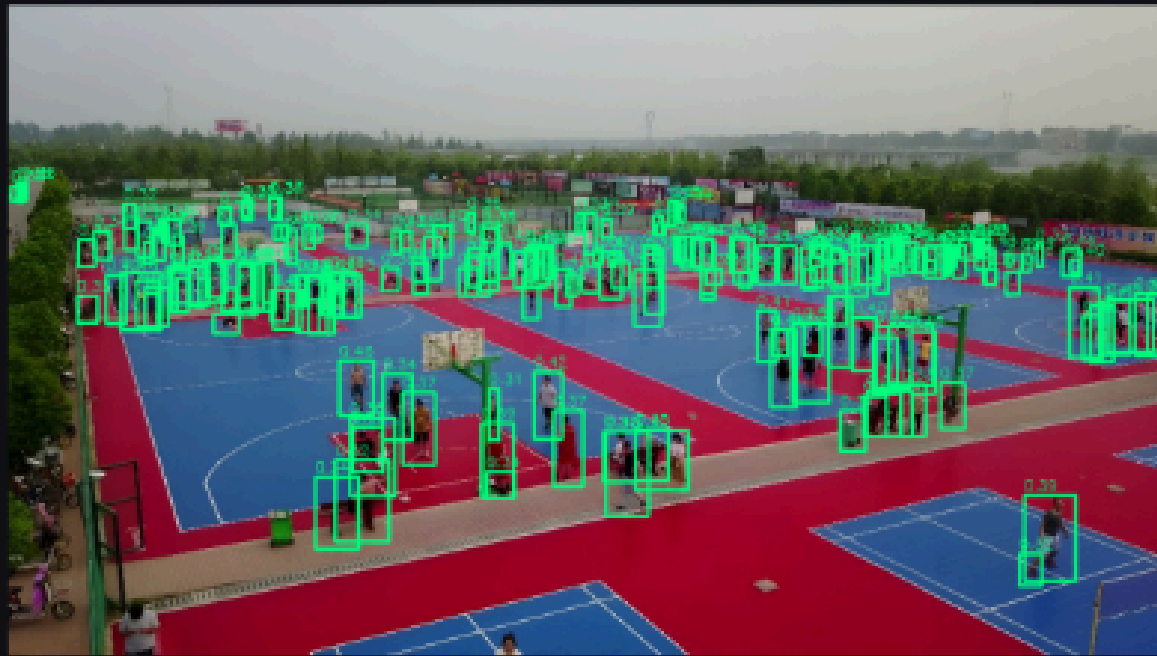
GT boxes : 2204
True positives: 602
Precision : 0.3460
Recall : 0.2731
mAP@50 (proxy): 0.3053

COMPLETE RESULTS – ALL 6 CONDITIONS

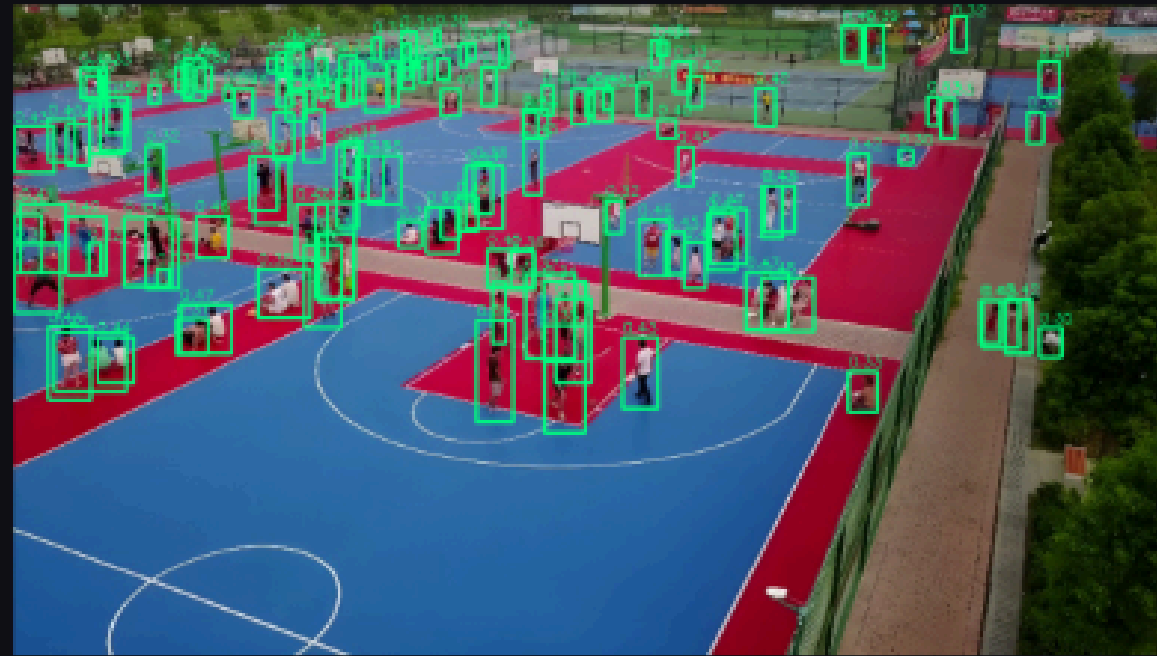
Condition	mAP@50	mAP50-95	Precision	Recall	Δ mAP@50
Clean (8× Zoom)	0.0588	0.0221	0.2405	0.2871	baseline
Blur Only	0.0595	0.0222	0.2346	0.2923	+0.0007
Noise Only	0.0509	0.0192	0.2426	0.2549	-0.0079
Blur+Noise	0.0503	0.0190	0.2420	0.2389	-0.0085
ESRGAN Enhanced	0.0601	0.0221	0.2360	0.2909	+0.0013
SAHI + ESRGAN	0.3053	0.1679	0.3460	0.2731	+0.2465

SAHI + ESRGAN — Top 3 Images by Detection Count

0000081_00000_d_0000001.jpg
192 detections



0000086_01954_d_0000005.jpg
125 detections



0000086_01443_d_0000004.jpg
118 detections



RT-DETR · VisDrone Tiny Object Detection
Clean vs Blur vs Noise vs Blur+Noise vs ESRGAN vs SAHI+ESRGAN

